Baselines First

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Responsible Plant Classifier

WORKBOOK · 1 OF 21

Recall and choose your starting point

> Record your starting explanation for Baselines First before opening notes. Show any later correction separately.

Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

Dataset Cards

Without opening the old lesson, explain one key idea from Dataset Cards. Give an example and a mistake that the idea helps you avoid.

> **My remembered idea, example and correction**

Sampling and Intended Use

Without opening the old lesson, explain one key idea from Sampling and Intended Use. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Python and Table Readiness

Without opening the old lesson, explain one key idea from Python and Table Readiness. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

WORKBOOK · 2 OF 21

Practise with fading help

> Complete the supported Baselines First trace and record the first step you can explain without help.

Use W2: Calculate the validation score for A **baseline** before a model. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs	Procedure I chose	Middle steps	Result and check

The step I completed without help, and the hint I needed

WORKBOOK · 3 OF 21

Individual reasoning check

> Check the proposed Baselines First explanation against the supplied facts before discussing it.

Use the worked case for A **baseline** before a model. Proposed explanation: “A simple **baseline** does not need **training** data”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

My explanation before partner discussion

WORKBOOK · 4 OF 21

Independent transfer case

> Write one first attempt at the reserved independent case before feedback. Repeated copies of that case refer to the same recorded attempt.

Independent case: Fit and compare independently

New **training** labels are blocked, clear, blocked, blocked, clear. New validation labels are clear, blocked, clear, blocked. Fit the same **baseline** rule and list its four **prediction**s.

A supplied candidate predicts clear, blocked, blocked, blocked. Score both on these rows and state the difference in correct examples and **percentage point**s. What claim remains unsupported?

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

WORKBOOK · 5 OF 21

Your learning target

> Connect the Baselines First target to a trace and a justified conclusion.

A dataset contains 27 healthy and 3 spotted leaves. An always-healthy **baseline** gets what **accuracy** and spotted recall? What must a new model improve upon?

WORKBOOK · 6 OF 21

Solve the given case

> Solve the supplied Baselines First case and show the deciding evidence.

A test set has eighteen healthy and two spotted leaves. **Baseline** always predicts healthy.

Calculate and critique **accuracy**.

Show your trace, calculation or **evidence**

> **Write or draw your reasoning and evidence.**

Check a changed case

> Apply the procedure to the changed case and record why the conclusion follows.

A dataset has 18 grass and 2 fern examples. A **baseline** always predicts grass. Compute **accuracy** and name a serious limitation.

My independent answer and reason

> **Write or draw your reasoning and evidence.**

WORKBOOK · 8 OF 21

Design a revealing test

> Specify a changed input, expected result and observation separately.

Create a changed **input**, boundary case or missing-information case. Write the expected answer or safe response before testing.

Test record	My evidence
Changed input or condition	
Expected answer or fallback	
Observed result	
Why the result occurred	

WORKBOOK · 9 OF 21

Project checkpoint

> Justify what Baselines First adds to Responsible Plant Classifier and keep the evidence.

Record majority and transparent-**rule baseline**s for the Plant Classifier using the frozen test plan.

The project change and **evidence** I will keep

Write or draw your reasoning and evidence.

WORKBOOK · 10 OF 21

Self check

> Judge explanation, evidence, transfer and limits separately; attach the work behind your judgement.

- I can explain the mechanism.
- I can show evidence.
- I can handle a changed case.
- I can name a limit.

Mark each as not yet, with help or independently, and attach supporting work.

Teacher review _____ Next step _____

WORKBOOK · 11 OF 21

Transfer practice from the original programme

> Use this earlier case as transfer practice. A case already discussed with feedback is not fresh assessment evidence.

A dataset contains 27 healthy and 3 spotted leaves. An always-healthy **baseline** gets what **accuracy** and spotted recall? What must a new model improve upon?

My independently reasoned answer

Record support used: none, hint or teacher explanation.

W1 Choose from training evidence

Choose from training evidence. Apply the supplied Baselines First facts and keep your reasoning.

Count clear and blocked in the worked **training** list. State the constant **prediction** and the declared tie rule. Explain why the validation majority does not choose this **baseline**.

My reasoning and evidence

W2 Calculate the validation score

> Calculate the validation score. Apply the supplied Baselines First facts and keep your reasoning.

List the four **baseline prediction**s and mark which validation rows are correct. Give correct/total and **accuracy**. Explain why 4/6 is not the answer.

My reasoning and evidence

W3 Make the comparison fair

> Make the comparison fair. Apply the supplied Baselines First facts and keep your reasoning.

A candidate is scored on ten easy rows and the **baseline** on four difficult rows. Describe a fair comparison and two details you must record.

My reasoning and evidence

W4 Fit and compare independently

> Attempt before feedback. If this repeats the selected independent case, reuse that first attempt; otherwise keep a separate record. Fit and compare independently. Apply the supplied Baselines First facts and keep your reasoning.

New **training** labels are blocked, clear, blocked, blocked, clear. New validation labels are clear, blocked, clear, blocked. Fit the same **baseline** rule and list its four **prediction**s.

A supplied candidate predicts clear, blocked, blocked, blocked. Score both on these rows and state the difference in correct examples and **percentage point**s. What claim remains unsupported?

> **My reasoning and evidence**

W5 Explain the hidden error

> Explain the hidden error. Apply the supplied Baselines First facts and keep your reasoning.

A **training**-derived clear **baseline** gets nine of ten validation cases correct; the labels are nine clear and one blocked. State its **accuracy** and missed case. If a candidate also gets nine correct, must it miss the same case?

Explain.

My reasoning and evidence

WORKBOOK · 17 OF 21

Feedback, return check and learning record

> Record the first repaired step and a different case to revisit after a delay.

Evidence	My record
Worked alone / hint / model	
First step I repaired and why	
New case checked later; date and result	
What I can now explain without notes	
One remaining question	

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

WORKBOOK · 18 OF 21

Pictorial case practice: Baselines First

> Make a simple baseline earn its place before claiming a better classifier. Explain the labelled

> step before checking the changed condition.

Make a simple **baseline** earn its place before claiming a better classifier.

Training labels: healthy, healthy, spotted. Majority **baseline** predicts healthy. Validation references: healthy, spotted, healthy, spotted.

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

AI ACADEMY / AI-4/10

Baselines First

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Fit baseline

Training majority is healthy: 2 of 3 labels.



2. Score baseline

Always healthy matches 2/4 validation cases: 50%.



3. Compare candidate

Supplied candidate predictions healthy, spotted, spotted, spotted match 3/4: 75%.

CHANGE ONE THING

Change the third candidate prediction from spotted to healthy.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Can the candidate's **training accuracy** replace this validation comparison? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

My picture-based explanation and changed-case prediction

Record the help used

Support used: alone / hint / worked example. This record is practice **evidence**.

Project connection: For your Responsible Plant Classifier, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

WORKBOOK · 19 OF 21

Project portfolio milestone

> Keep this week's input, deciding step, result and limit together in the Responsible Plant Classifier record.

Project: Responsible Plant Classifier

Current milestone: Make a transparent first version. Keep a trace of how one **input** becomes an **output**; explain the rule or learned component.

For your Responsible Plant Classifier, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

Evidence to keep	My record
This week's input and deciding step	
Prediction and actual result (or labelled paper trace)	
One change since the previous checkpoint and why	

Evidence to keep	My record
One error or limit and the next test	

> **Explain what this week adds to your project. Reuse last week's record rather than starting again.**

WORKBOOK · 20 OF 21

Supported practice and learning evidence

> Try the supported questions before comparing with hints and feedback. Record what you repaired and the evidence you retained.

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Make a simple **baseline** earn its place before claiming a better classifier.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? Supplied candidate **prediction**s healthy, spotted, spotted, spotted match 3/4: 75%.

1. Fit **baseline**

2. Score **baseline**

3. Compare candidate

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Compare candidate Supplied candidate **prediction**s healthy, spotted, spotted, spotted match 3/4: 75%. Read the picture in order.

Supplied candidate **prediction**s healthy, spotted, spotted, spotted match 3/4: 75%. The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Practice 2: explain the deciding evidence

Changed case · predict first

Changed-case practice: Change the third candidate **prediction** from spotted to healthy. Which conclusion is supported by the supplied facts?

1. Candidate becomes correct on all 4 validation cases.

2. We can decide the result without examining the changed **input** or the procedure.

3. This one example guarantees correct results for every future **input**.

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Candidate becomes correct on all 4 validation cases. Only the previously wrong third decision changes; **baseline** stays fixed at 2/4.

Repair a missing building block

Review Dataset Cards (ai-4/9). Goal: Inspect row meaning missing values and repeated records. Short practice: What two counts should the revised card show?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Sampling and Intended Use (ai-4/7). Goal: Inspect row meaning missing values and repeated records. Short practice: Which collection would address the intended-use gap most directly?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Python and Table Readiness (ai-4/4). Goal: Specify a table schema. Short practice: Why should the missing value not silently become zero?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Predict, run and debug

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual **output** or error, change one **input**, then run again.

Keep code, **prediction**s, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

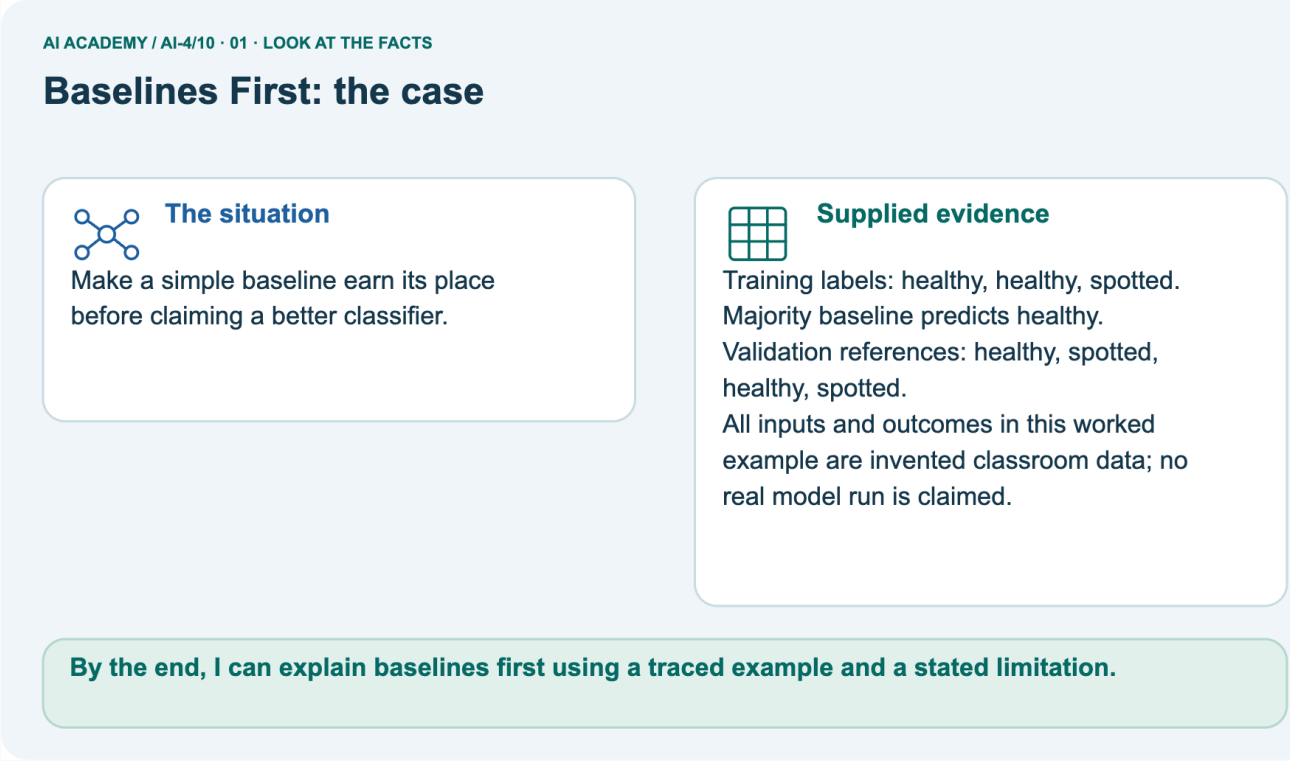
What I repaired, the evidence I kept and the next test

WORKBOOK · 21 OF 21

Visual concept journey

> Explain each Baselines First picture from its labels, then record a relevant limit.

01. Baselines First: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

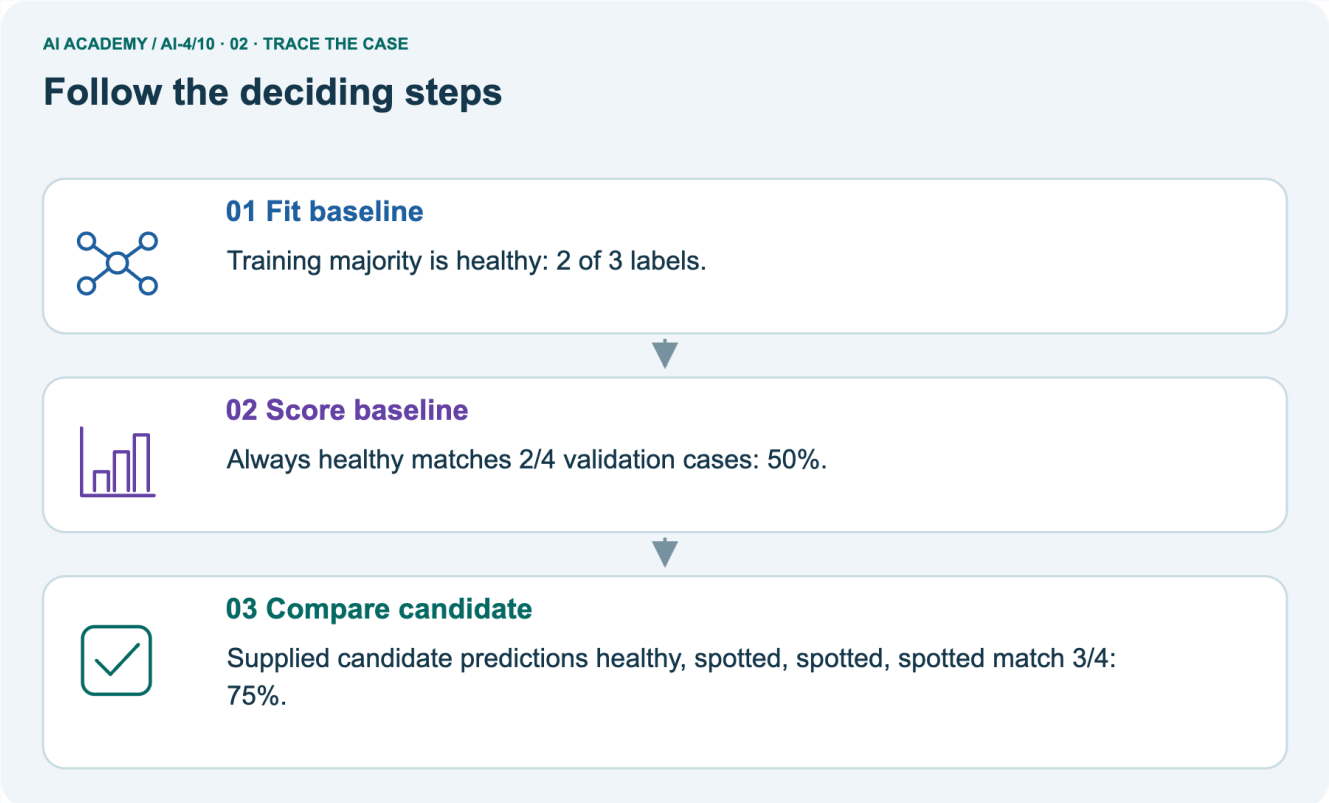
Make a simple **baseline** earn its place before claiming a better classifier. Supplied facts: **Training** labels: healthy, healthy, spotted. Majority **baseline** predicts healthy.

Validation references: healthy, spotted, healthy, spotted. All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

> Point to the given fact that matters most. Explain why it matters.

My picture explanation, deciding evidence and one limit

02. Follow the deciding steps



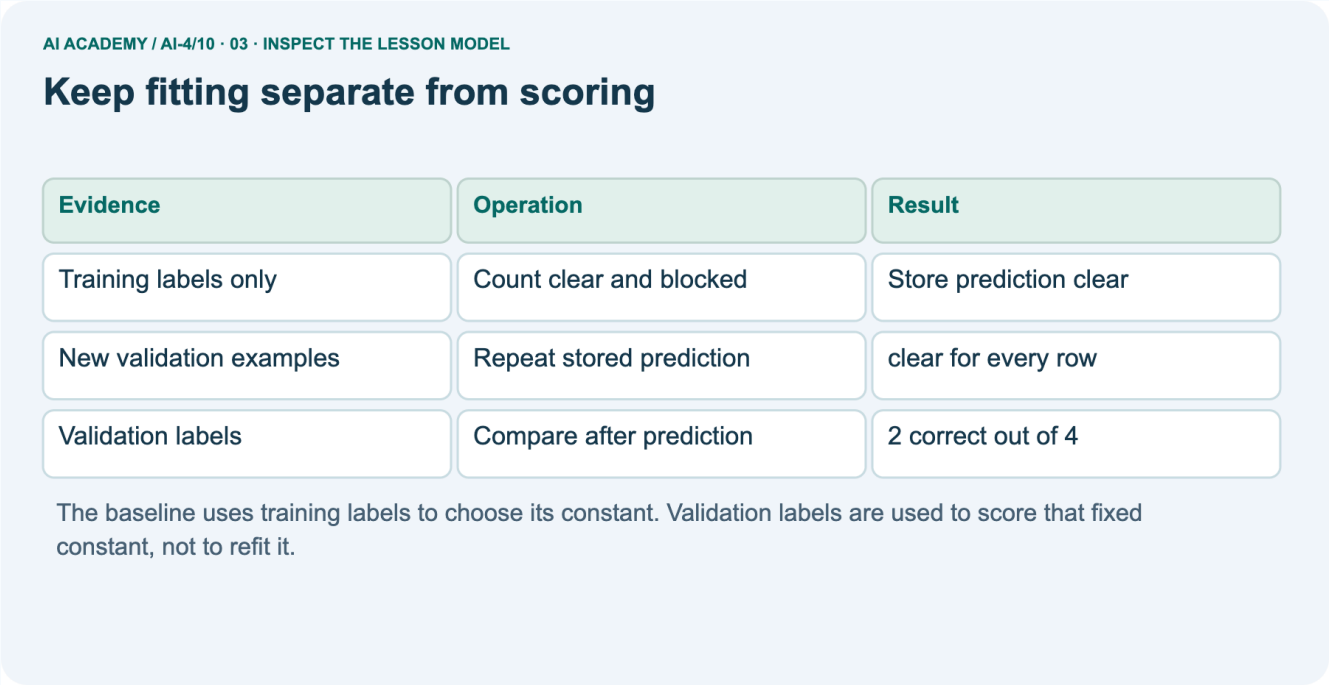
Follow the labelled picture with the matching explanation. Open the vector diagram

Fit **baseline**: **Training** majority is healthy: 2 of 3 labels. → Score **baseline**: Always healthy matches 2/4 validation cases: 50%. → Compare candidate: Supplied candidate **prediction**s healthy, spotted, spotted, spotted match 3/4: 75%.

Point to each numbered stage and explain which recorded **input** it uses.

My picture explanation, deciding evidence and one limit

03. Keep fitting separate from scoring



Follow the labelled picture with the matching explanation. Open the vector diagram

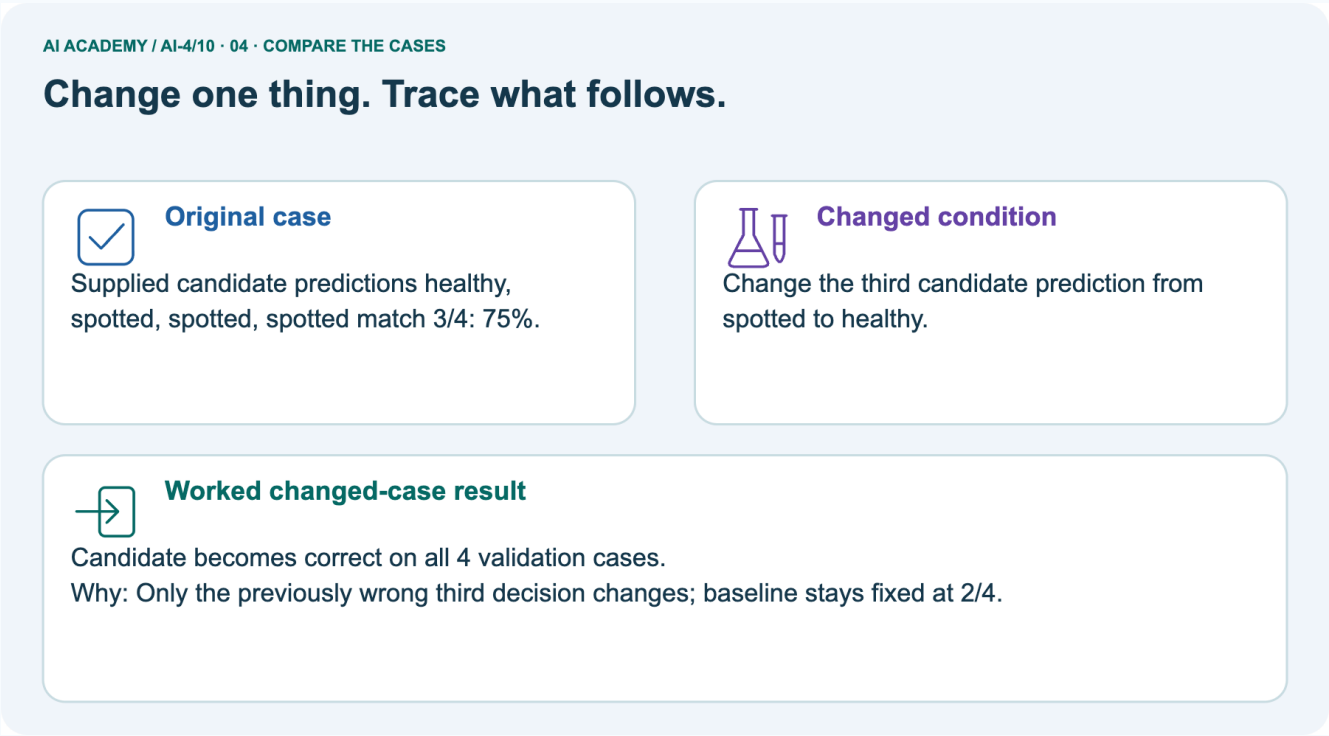
Keep **fitting** separate from scoring. **Training** labels only | Count clear and blocked | Store **prediction** clear. New validation examples | Repeat stored **prediction** | clear for every row.

Validation labels | Compare after **prediction** | 2 correct out of 4

Read the labels and values. Explain the relationship the model shows.

My picture explanation, deciding evidence and one limit

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

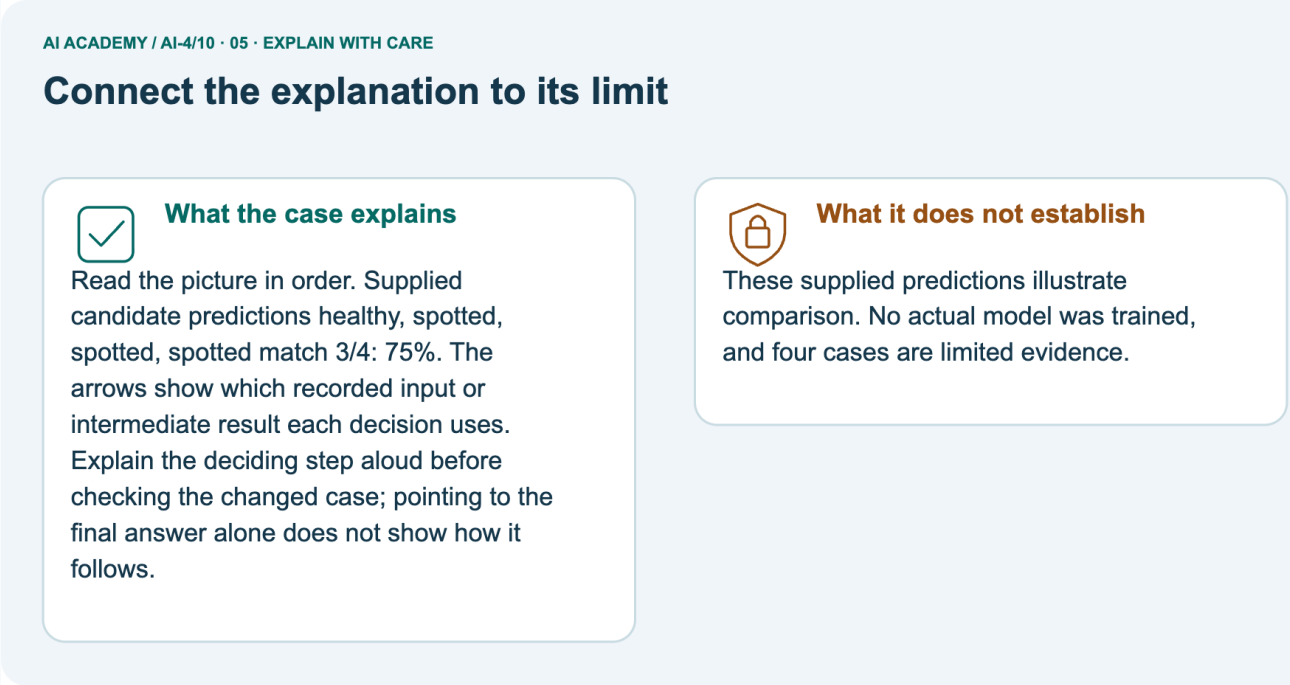
Original case: Supplied candidate **prediction**s healthy, spotted, spotted, spotted match 3/4: 75%.
Changed condition: Change the third candidate **prediction** from spotted to healthy.

Expected result: Candidate becomes correct on all 4 validation cases. Reason: Only the previously wrong third decision changes; **baseline** stays fixed at 2/4.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

My picture explanation, deciding evidence and one limit

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture in order. Supplied candidate **prediction**s healthy, spotted, spotted, spotted match 3/4: 75%. The arrows show which recorded **input** or intermediate result each decision uses.

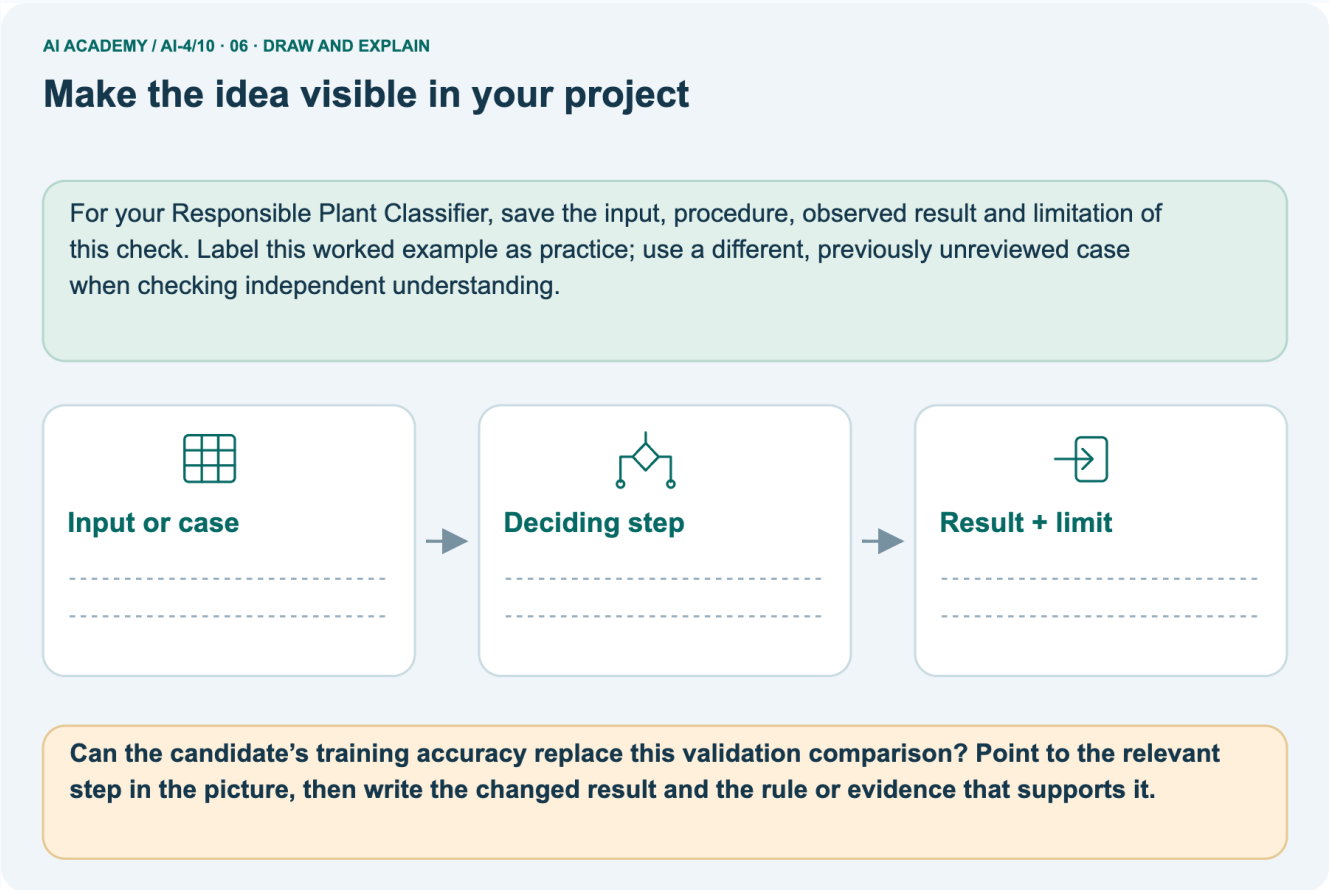
Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows. Limit: These supplied **prediction**s illustrate comparison.

No actual model was trained, and four cases are limited **evidence**.

Explain the case, then point to a limit on the claim.

My picture explanation, deciding evidence and one limit

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For your Responsible Plant Classifier, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding. Draw your **input**, deciding step, result and limit.

Supported question: Can the candidate's **training accuracy** replace this validation comparison? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

> Draw your project trace and justify the deciding step using **evidence**.

My picture explanation, deciding evidence and one limit

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.